

Aditya Vasu Rana

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Education

Vellore Institute of Technology, Chennai

Aug 2023 – May 2027

Bachelor of Technology (B.Tech) in Computer Science, **CGPA : 9.5 (Branch Rank 3)**

- **Coursework:** Computer Networks, Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Database Management Systems

Technical Skills

- **Programming Languages:** Python, R, C, C++ ,Java
- **Data Analysis & Modeling:** Machine Learning, Regression, Classification, Clustering, Neural Networks, CNN
- **Tools & Platforms:** Git, AWS, Docker, Jupyter, Streamlit , Hex
- **Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Matplotlib, Seaborn
- **Databases & Visualization:** SQL, PL/SQL, Tableau, Power BI, Data Storytelling
- **Web Technologies:** HTML, CSS, JavaScript, ReactJS, Flask, FastAPI, Tailwind, Bootstrap

Projects

Fashion Recommender System - CNN + ResNet50 + KNN

github.com

- Built a **content-based fashion recommendation system** using **ResNet50 (ImageNet pretrained)** as a feature extractor to generate **2048-D deep embeddings** for 44k fashion product images.
- Implemented an efficient image retrieval pipeline using **GlobalMaxPooling, L2 normalization**, and **K-Nearest Neighbors (KNN)** to recommend the **Top-5 visually similar items**.
- Developed an interactive **Streamlit web application** that allows users to upload a fashion image and receive real-time recommendations using precomputed embeddings stored in `embeddings.pkl`.

Potato Leaf Disease Classification - Deep Learning + Flask Backend

github.com

- Built a **Potato Leaf Disease Classifier** on the PlantVillage dataset (**1.5K images, 3 classes**); baseline CNN using 4 convolutional blocks and 1 dense layer ,showed overfitting (train **95%**, val **35–50%**).
- Improved generalization using **MobileNetV2 transfer learning** with augmentation, batch normalization, dropout (0.4), and early stopping, achieving **95% validation accuracy**.
- Deployed the pipeline as a **Flask-based web app** with minimal frontend for real-time prediction.

Customer Churn Prediction - Machine Learning + Flask Backend

github.com

- Built an end-to-end customer churn prediction system using Logistic Regression and Random Forest on telecom customer data, achieving up to **0.78 accuracy** and **0.72 recall for churn class**.
- Implemented a robust ML pipeline with EDA, preprocessing, class-imbalance handling (SMOTE and class weighting), and threshold tuning to improve minority-class detection.
- Developed an interactive **Flask-based web application** with SHAP-powered explainability to surface top risk drivers and enable real-time churn risk assessment with dynamic churn probability threshold tuning.

Experience

Data Science and Analytics Intern, Kerala Ayurveda Ltd. (Hybrid)

Oct 2025 – Jan 2026

- Analyzed D2C sales funnel across multiple product categories using Python and SQL, identifying key drop-offs such as low Add-to-Cart (ATC) and Cart-to-Checkout (C2C) rates for sales improvement initiatives.
- Conducted customer behavior analysis including low ATC diagnosis, abandoned cart AOV analysis, and first-product to next-likely-product mapping, generating actionable insights to optimize conversion journeys.
- Collaborated closely with the product management team to frame hypotheses, validate marketing strategies, and support data-driven decisions aimed at improving sales performance and customer engagement.

Board Member, E-CELL VIT Chennai

Apr 2025 – Present

- Organized 5+ flagship events/workshops, boosting participation by 40%, and managed sponsorships, partnerships, and cross-functional teams to strengthen the startup ecosystem.

Certifications and Achievements

- **Oracle Cloud Infrastructure Architecture: Data Science Professional**, Oracle University oracle.com
- NDA AIR 4 (2022-1) – Secured All India Rank 4 in NDA entrance exam.
- 2nd place at V-Medithon Intercollege Hackathon , Ranked 1 for ML model.
- 300 problems solved on LeetCode with 100 days badge leetcode.com